

Project Pong

A modern recreation of the classic **Pong** game built in **C++** using **raylib**.

This project started as a way to learn raylib by following Programming With Nick's tutorials, but quickly grew into something much bigger. Instead of stopping at a simple ball-and-paddle demo, I expanded it into a complete local multiplayer game with menus, settings, music, sound effects, pause functionality, and customizable gameplay.

It's a small project, but it taught me a lot about game loops, collision detection, audio management, state machines, and organizing game logic in C++.

Features

Gameplay

- 🏓 Local **2-player Pong**
- ⚡ Ball speed increases after every paddle hit to keep matches exciting
- 🏆 Choose the winning score before the match:
 - 5 Points
 - 10 Points
 - 15 Points

Menus

- Main Menu
- Settings Menu
- Pause Menu
- Game Over Screen
- Play Again / Return to Main Menu

Audio

- 🎵 Choose from **5 different background music tracks**
- 🚫 Option to turn background music off
- 🔊 Toggle sound effects on or off independently

Game Over

- Displays the winner
 - Shows the final score
 - Restart the game instantly or return to the main menu
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Controls

Player	Controls
Left Paddle	W (Up) / S (Down)
Right Paddle	↑ / ↓ Arrow Keys
Pause / Resume	ESC

Built With

- C++17
- raylib

Project Structure

```
Project-Pong/  
|  
├─ main.cpp  
├─ *.mp3  
├─ README.md  
└─ ...
```

Files

File	Description
<code>main.cpp</code>	Contains the complete game logic
<code>*.mp3</code>	Background music and sound effects

How to Run

1. Install raylib

Install raylib for your operating system.

2. Compile

```
g++ -std=c++17 main.cpp -o main -lraylib
```

3. Run

```
./main
```

macOS / Linux Notes

Depending on your operating system, raylib may require additional linker flags.

For example:

- macOS requires frameworks such as Cocoa and OpenGL.
- Linux often requires X11, pthread, GL, dl, and rt libraries.

If compilation fails during linking, refer to the raylib installation guide for your operating system.



Audio Files

The game loads all music and sound effects using **relative file paths**.

Make sure all `.mp3` files remain in the same directory as the executable (or update the paths in the source code). Missing audio files may prevent sounds from loading correctly.



What I Learned

Building this project helped me gain practical experience with:

- Game loops
 - Collision detection
 - Game state management
 - Input handling
 - Audio playback
 - Menu systems
 - Object-oriented programming in C++
 - Working with the raylib game framework
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Future Improvements

- 🎮 Single-player mode with an AI opponent
- 💾 Save high scores
- 🌐 Online multiplayer

- 🎮 Improved graphics and animations
 - 🎵 Better sound effects (the current paddle hit is a fart sound... it gets the job done 😊)
 - 🛠 Split the project into multiple source/header files instead of keeping everything in `main.cpp`
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🙏 Credits

This project was originally inspired by the excellent **Programming With Nick** raylib tutorials.

After completing the tutorial, I expanded the project with my own features including:

- Main menu
 - Settings menu
 - Pause system
 - Multiple music tracks
 - Audio toggles
 - Game over screen
 - Winning score selection
 - Increasing ball speed
 - General gameplay improvements
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★ About

This project was built as part of my journey learning **C++** and **raylib**.

It may be a simple game, but every feature added was an opportunity to learn something new. I'll continue improving it whenever I have time and use it as a foundation for building larger game projects in the future.

If you have suggestions or feedback, feel free to open an issue or fork the project.

Thanks for checking it out!